Federico Tesone

Principal Engineer · Search, Retrieval & Agentic AI Systems

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Summary

Principal Software Engineer with 14+ years building large-scale search, retrieval, recommendation, and distributed systems. Leads Search Core at Daydream, architecting the platform behind an AI-native conversational search product: hybrid retrieval across 5M+ product options, LLM query understanding running as a concurrent DAG with confidence routing, multi-intent splitting, and agentic fallback loops, multi-turn preference memory, an offline/online evaluation platform that gates ranking and prompt changes, and end-to-end tracing of LLM decisions. Previously led technical strategy and architecture for Grubhub's search and discovery platform at 120K–150K requests per minute — modular search runtime, +2.37% search conversion, −34% Elasticsearch CPU. Experienced in defining architecture, building production ML systems, and leading initiatives across engineering, product, and machine-learning teams.

Experience

Principal Engineer, Daydream – New York, NY, USA July 2025 – present

Lead the Search Core team owning retrieval, ranking, and query understanding for an AI-native fashion search and conversational engine.

- Architected a search engine from the ground up, advancing from lexical retrieval to a hybrid system fusing text, visual, and semantic signals into a single ranked result set across a catalog of 5M+ searchable product options.
- Re-architected query understanding from a single blocking LLM call into a concurrent DAG with confidence routing and multi-intent splitting — cut P50 latency ∼18% and P95 ∼20% versus the legacy path, despite substantially more signal extraction per query.
- Designed multi-pass re-ranking and LLM-driven query understanding with agentic fallback loops.
- Extended the platform to image search, visual discovery, and multi-turn preference memory.
- Built an offline/online evaluation platform for gating ranking and prompt changes, plus embedding pipelines for personalization.
- Developed end-to-end query tracing and observability tooling to inspect LLM decisions, retrieval stages, ranking signals, latency, and final results.

Staff Engineer, Grubhub – New York, NY, USA Feb 2023 – July 2025

Led technical strategy and architecture for Grubhub's high-throughput search and discovery platform.

- Architected a modular search runtime separating context, candidate selection, ranking, experimentation, and hydration, enabling independent evolution of retrieval and ML ranking systems.
- Operated and evolved Grubhub's search and discovery platform serving approximately 120K–150K requests per minute across major consumer surfaces.
- Improved search conversion by 2.37% through multi-query retrieval and result blending.
- Reduced Elasticsearch CPU utilization by 34% by redesigning retrieval from multi-query execution to a single-query architecture, while improving ranking flexibility and production scalability.
- Designed candidate selection and ranking infrastructure integrating real-time ML model predictions, embeddings, business constraints, geographic filtering, and experimentation signals.
- Defined technical strategy across Search, Recommender Systems, Embeddings, and related discovery infrastructure; mentored senior engineers across multiple backend and search initiatives.

Senior Software Engineer, Grubhub – New York, NY, USA Dec 2019 – Feb 2023

Engineered high-availability backend microservices for the discovery runtime platform.

- Built a low-latency vector retrieval system for production runtime workloads using Annoy, with custom sharding of vector indexes to support scalable embedding-based lookup across large datasets.
- Developed low-latency data serving infrastructure for embedding-based ML models and user targeting.
- Partnered with machine-learning and data teams to productionize new ranking and recommendation capabilities.

Senior Developer, Endava – Rosario, Argentina July 2016 – Dec 2019

Developed backend systems and microservices using Java, Scala, Python, Cassandra, AWS, and CI/CD pipelines.

Auxiliary Professor, Universidad Abierta Interamericana – Rosario, Argentina Aug 2014 – Mar 2019

Taught Object-Oriented Software Design, GoF Design Patterns, Software Metrics, QA, and Automated Testing.

Team Lead & Senior Developer, Nación Servicios – Rosario, Argentina July 2013 – July 2016

Led development of core banking loyalty platforms, transaction authorization systems, and EMV payment services.

BI Consultant, BPD Solutions – Rosario, Argentina Mar 2012 – June 2013

Engineered enterprise ETL pipelines, data integration workflows, and data warehousing systems.

Education

Universidad Tecnológica Nacional, MSc in Software Engineering 2013 – Nov 2021

Thesis: Developing a Similarity Measure for Recommender Systems on Community Question Answering Sites: A Big Data Analysis Using a Clustering Ensemble Method. Written in Spanish: *Desarrollo de una medida de similaridad para Sistemas de Recomendación en sitios de Community Question Answering. Análisis desde un enfoque Big Data y usando un método de ensamble de clustering.*

Universidad Tecnológica Nacional, BSc in System Information Engineering 2007 – 2012

5-year degree with minor thesis.

Technical Expertise

AI & Search Systems: Hybrid Retrieval, Semantic Search, Dense & Sparse Retrieval, Vector Search, Ranking, Re-ranking, LLM Query Understanding, Agentic Search, Multi-turn Preference Memory, Evaluation Systems, Embeddings, Personalization, Recommendation Systems

Backend & Data Infra: High-Throughput Services, Low-Latency Systems, Real-Time Data Serving, Feature Infrastructure, Search Infrastructure, Event-Driven Systems, Observability, Reliability Engineering

Languages & Tech: Java, Scala, Python, SQL, Elasticsearch, Cassandra, ScyllaDB, Redis, Vector Databases, Google Cloud, AWS

Languages: English (Full Professional), Spanish (Native)

Honors and Awards

Cuadro de Honor - Honor Roll

Published in the list of highest grade point averages of the Province of Santa Fe, Argentina.